

Multi-Objective Optimization for Inverse Kinematics of Robotic Arm with Joint Configuration

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March 22-23, 2025

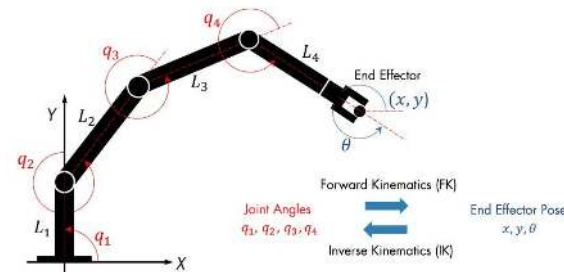
OUTLINE

- **Introduction and Background**
- **Methodology**
- **Algorithm Development**
- **Experiments and Results**
- **Conclusion and Future Work**

Introduction and Background

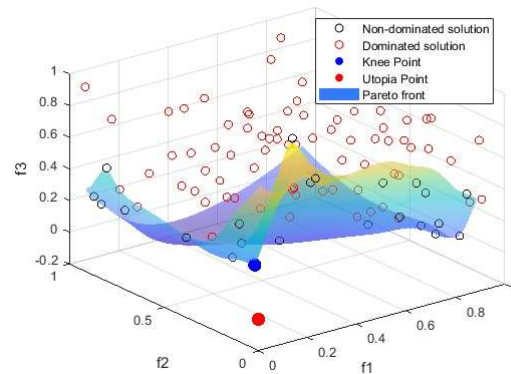
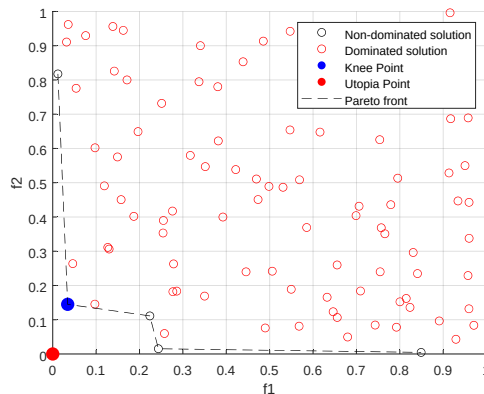
INTRODUCTION

- Collaborative robotic arms are widely used in manufacturing, healthcare, service industries, agriculture, and research labs for tasks like assembly, surgeries, customer service, harvesting, and experiments. They improve efficiency and precision through human-robot interaction. For example, in manufacturing, they can boost productivity by up to 40%.
- Inverse kinematics is important in these applications, ensuring precise control of the robotic arms' movements to achieve desired positions and orientations efficiently.



INTRODUCTION

- Many engineering problems can be seen as multi-objective optimization problems because they involve balancing several goals.
- Problem: Inverse kinematics of a robotic arm is to solve the joint rotation angles that enable the end effector to achieve a given target position and orientation. Inverse kinematics should be focused not only on calculating the joint rotations of the robot but also on minimizing energy consumption.



Background

➤ Motivations

- ◆ Reducing energy consumption of robotic arm operations.
- ◆ Extending the lifespan of the motors.
- ◆ Balancing multiple goals, general solution, reusable for different kinematic models.



Background

➤ Objectives

- ◆ Develop optimization algorithm for inverse kinematics, incorporating joint configuration of robotic arm through multi-objective optimization.
- ◆ Conduct experiments to validate the developed algorithm, test in both simulation and real robotic arm.

Methodology

- Forward kinematics
- Inverse kinematics
- Workspace analysis

Methodology

➤ Forward kinematics analysis

◆ DH parameters (standard method)

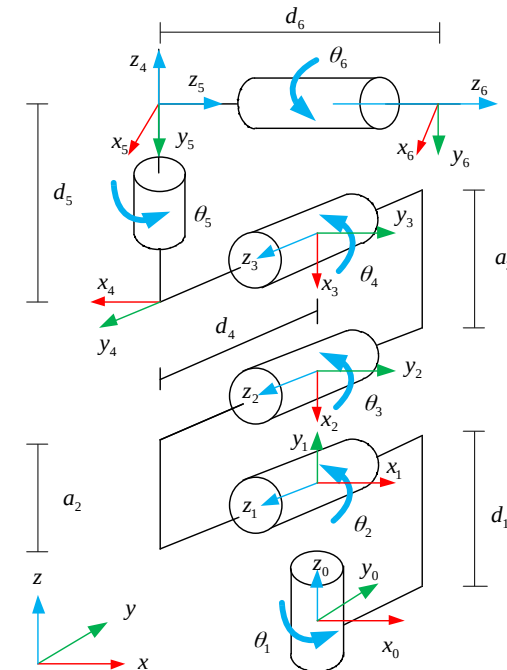
#	θ (rad)	d	a	α
0-1	θ_1	d_1	0	$\pi/2$
1-2	$\theta_2 - \pi/2$	0	$-a_2$	0
2-3	θ_3	0	$-a_3$	0
3-4	$\theta_4 - \pi/2$	d_4	0	$\pi/2$
4-5	$\theta_5 + \pi/2$	d_5	0	$-\pi/2$
5-6	θ_6	d_6	0	0



myCobot 280 Pi



UR3



Methodology

➤ Inverse kinematics

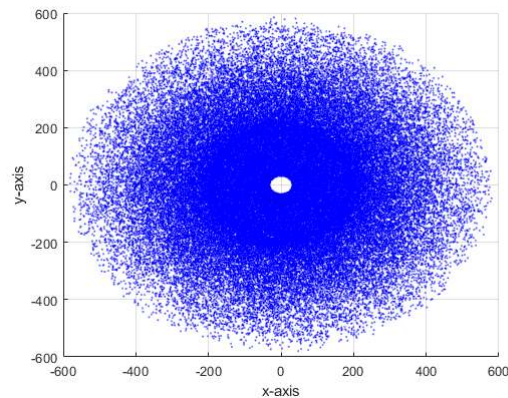
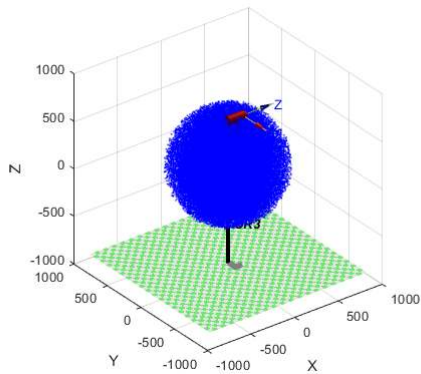
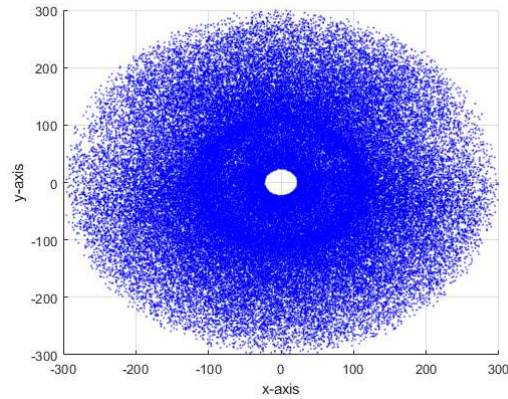
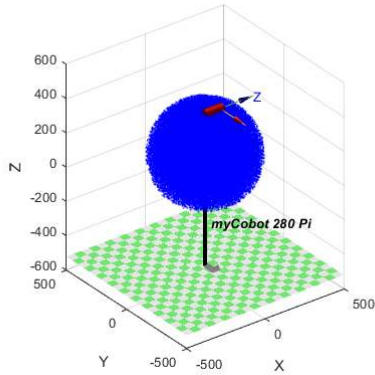
- ◆ For the optimization algorithms
 - ◆ Inputs: Target position, Target orientation (rotation matrix)
 - ◆ Outputs: $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$
- ◆ Constrain: Joint range limits

Model	myCobot 280 Pi (degrees)	UR3
Joint 1	-168~+168	-360~+360
Joint 2	-135~+135	-360~+360
Joint 3	-150~+150	-360~+360
Joint 4	-145~+145	-360~+360
Joint 5	-165~+165	-360~+360
Joint 6	-180~+180	-inf~+inf

Methodology

➤ Workspace analysis

◆ Monte Carlo method (in mm)



Algorithm development

Algorithms

➤ Particle swarm optimization

- ◆ 1. **Initialization:** Initialize a swarm of particles (population size H), including random positions and

velocities. $X_i = [x_{i1} \ x_{i2} \ x_{i3} \ x_{i4} \ x_{i5} \ x_{i6}]$ $x_{\min} < x_{ij} < x_{\max} \ j = 1, 2, \dots, 6$

$$V_i = [v_{i1} \ v_{i2} \ v_{i3} \ v_{i4} \ v_{i5} \ v_{i6}] \quad v_{ij} = \begin{cases} v_{\min} & \text{if } v_{ij} < v_{\min} \\ v_{ij} & \text{if } v_{\min} \leq v_{ij} \leq v_{\max} \\ v_{\max} & \text{if } v_{ij} > v_{\max} \end{cases} \quad j = 1, 2, \dots, 6 \quad [X_1 X_2 \dots X_H]^T$$

- ◆ 2. **Fitness evaluation:** Evaluate the fitness of each particle.
- ◆ 3. **Update individual best (pbest):** For each particle, compare its fitness value with the best position it has encountered (pbest). If the current value is better, update pbest to the current position.

$$pbest = [P_1 P_2 \dots P_H]^T$$

- ◆ 4. **Update global best (gbest):** For each particle, compare its fitness value with the global best position encountered by the entire swarm (gbest). If the current value is better, update gbest to the current position.

$$V_i = \omega V_i + c_1 r_1 (pbest_i - X_i) + c_2 r_2 (gbest - X_i)$$

Algorithms

➤ Particle swarm optimization

- ◆ 5. **Update velocity and position:** Update the velocity and position of each particle according to the function.
- ◆ 6. **Termination Check:** If the termination condition is not met (usually a sufficiently good fitness value or reaching a preset maximum number of generations G_{max}), return to step 2.

Algorithms

➤ Weighted multi-objective PSO (WMOPSO)



Algorithm WMOPSO for Inverse Kinematics

```
Set target position Pt and target orientation Ot
Initialize objective function values, weighs and solution stores
Initialize PSO parameters
Check joint range limitations
Iterate over weight combinations
for w1 = 0.001 to 0.005 step 0.0005 do
  for w3 = 0.001 to 0.005 step 0.0005 do
    w2=1-w1-w3
    if w2>0
      Execute PSO to obtain  $\theta_i$  ( $i = 1, \dots, 6$ )
      Calculate position matrix P and orientation matrix O,
      (P, O)
      Calculate f1, f2, f3
      if f1 and f2 <= allowable limits
        store objective function values, weighs and solution
      end if
    end if
  end for
end for
```

Experiments and Results

- Single points
- Trajectories

Experiments and Results

➤ Single points

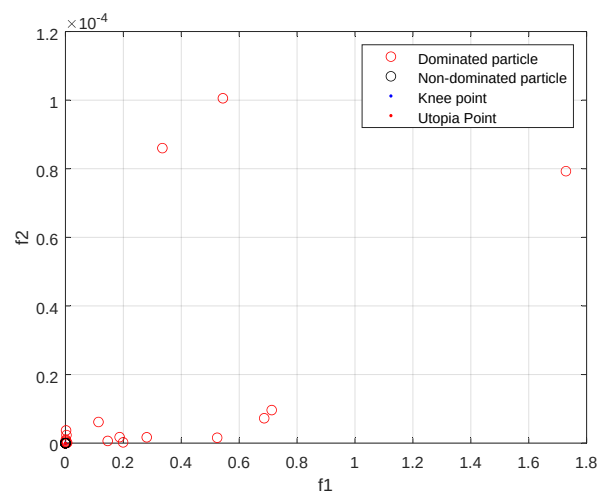
- ◆ Target position: (150, 150, 200) in millimeters
- ◆ Target orientation: End effector facing downward

Performance comparison of WMOPSO in single point experiments (average of 10 runs)

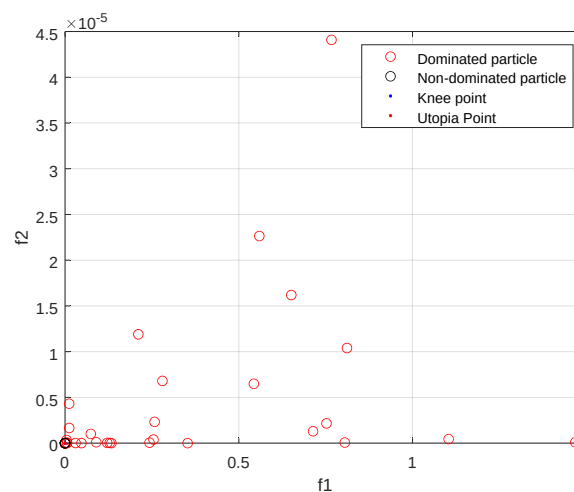
Algorithm	Position error (m)	Orientation error	Joint rotation mean (rad)
W2OPSO-mycobot280 (f1+f2)	0	1.3687×10^{-16}	5.5489
W2OPSO-UR3 (f1+f2)	0	3.1265×10^{-16}	4.8687
WMOPSO-mycobot280	0	1.5564×10^{-16}	3.1416
WMOPSO-UR3	0	3.1265×10^{-16}	4.8687

Experiments and Results

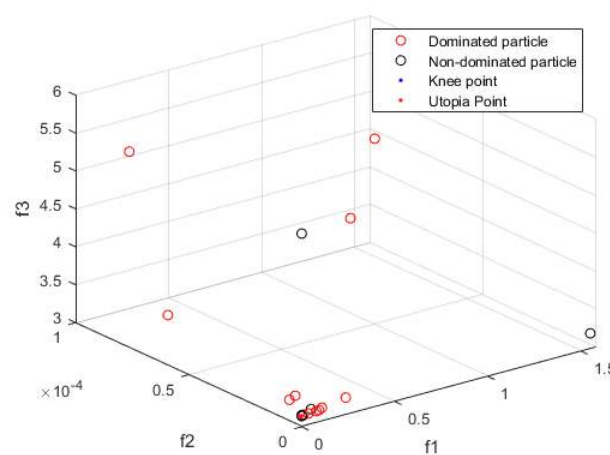
➤ Single points



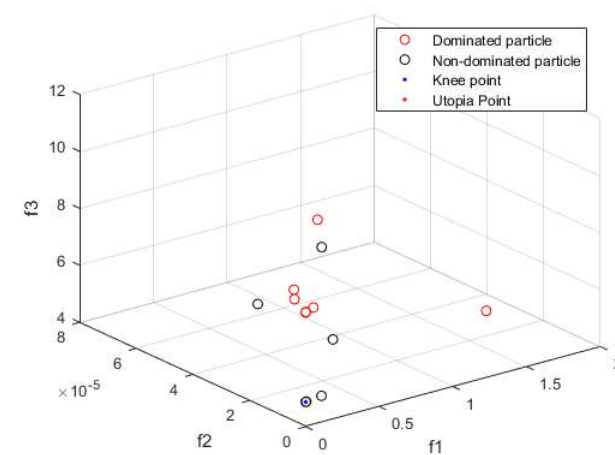
W2OPSO-mycobot280 (f1+f2)



W2OPSO-UR3 (f1+f2)



WMOPSO-mycobot280



WMOPSO-UR3

Experiments and Results

➤ Trajectories

◆ Trajectory information

Trajectory	Center/Start point (x, y, z) for <u>myCobot</u> 280 Pi (mm)	Center/Start point (x, y, z) for UR3 (mm)	Size	Number of Points
Circle	(100,100, 200)	(200,200, 200)	$r=50$ (mm)	$n=50$
Circle/	(150, 150, 200)	(200, 200, 200)	$r=50$ (mm)	$n=50$
Sine	(100, 100, 200)	(200, 200, 200)	$A = 100$ (mm) $f = 1/\pi$ $\phi = 2$ $T=100$	$n=50$

^a. NOTE: The “/” indicates the end effector with 45 degrees downward.

Experiments and Results

➤ Trajectories

◆ Results

Trajectory	Algorithm-Model	RSE of Position (m)	RSE of Orientation	Total Joint Rotations (rad)
Circle	WMOPSO-myCobot280	7.6068×10^{-8}	4.9398×10^{-8}	3.9735
	WMOPSO-UR3	5.1787×10^{-18}	1.9430×10^{-16}	4.3980
Circle/	WMOPSO-myCobot280	6.9991×10^{-8}	1.6162×10^{-8}	5.0374
	WMOPSO-UR3	5.1159×10^{-18}	2.3007×10^{-16}	6.1968
Sine	WMOPSO-myCobot280	1.7547×10^{-13}	1.1854×10^{-13}	4.9485
	WMOPSO-UR3	4.7474×10^{-18}	2.0211×10^{-16}	5.1709

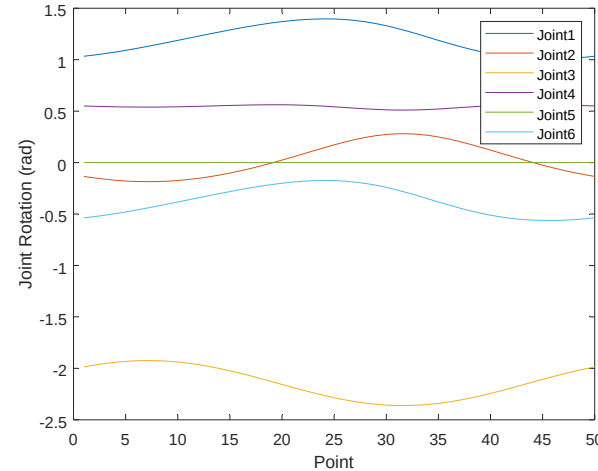
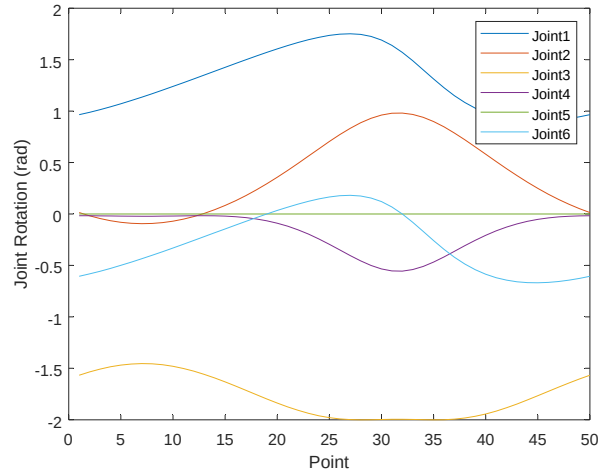
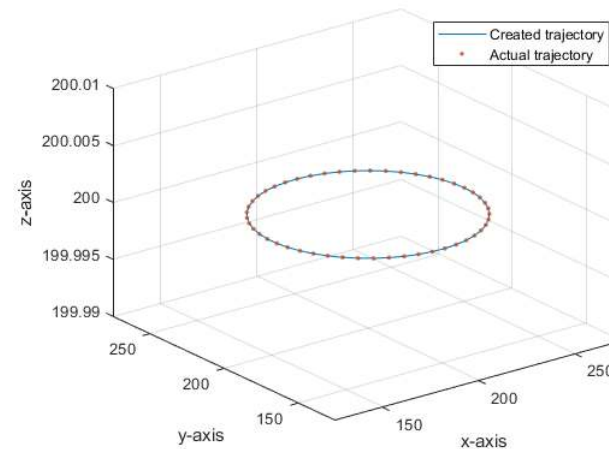
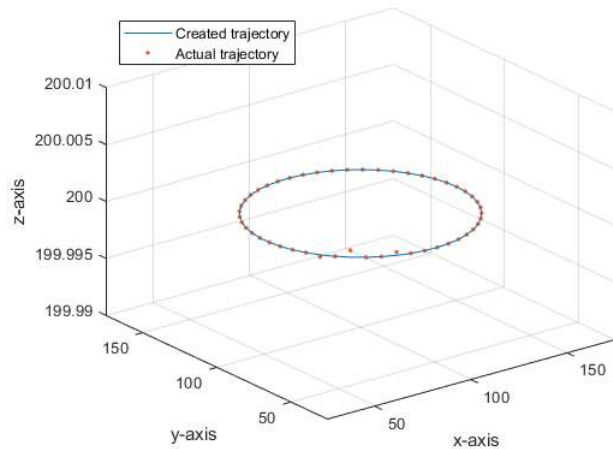
- ◆ Performance comparison for circle trajectory (50 points)

Algorithm	Position error (m)	Orientation error
MO-M-PSO (Best)	2.31×10^{-6}	1.85×10^{-8}
WMOPSO (Mean)	3.8034×10^{-8}	2.4699×10^{-8}

Experiments and Results

➤ Trajectories

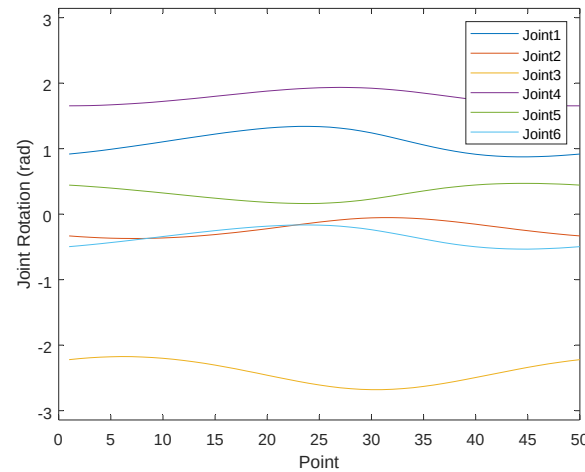
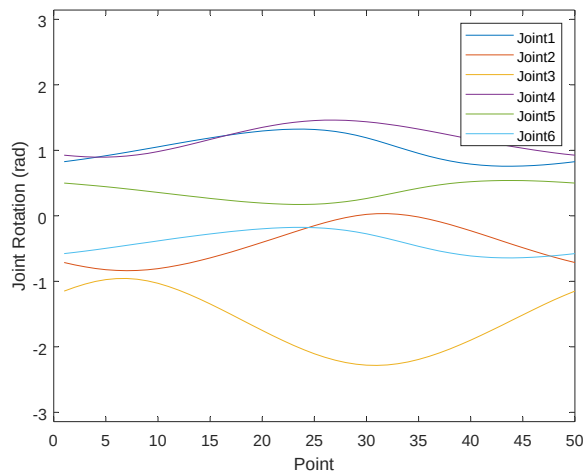
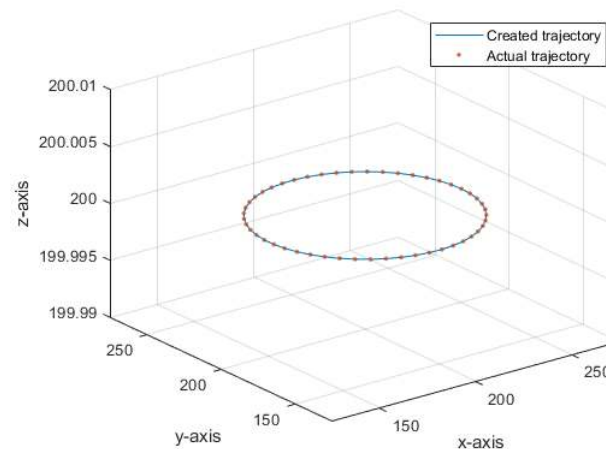
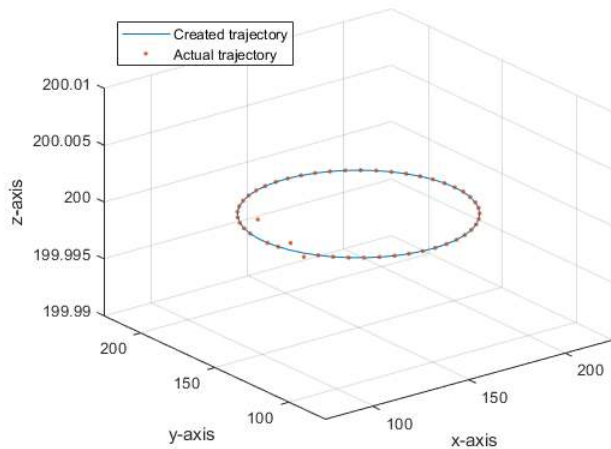
- ◆ Results for circle trajectory with gripper downward



Experiments and Results

➤ Trajectories

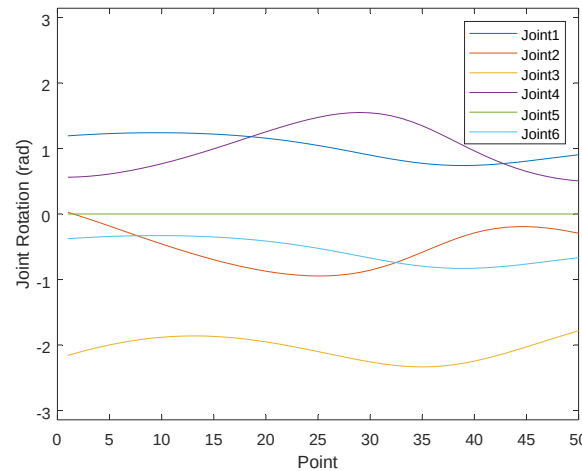
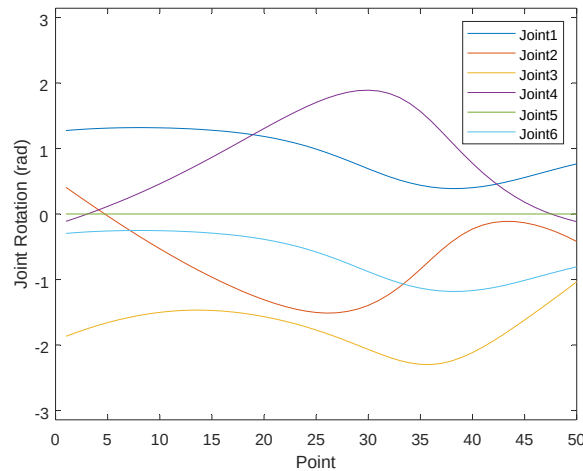
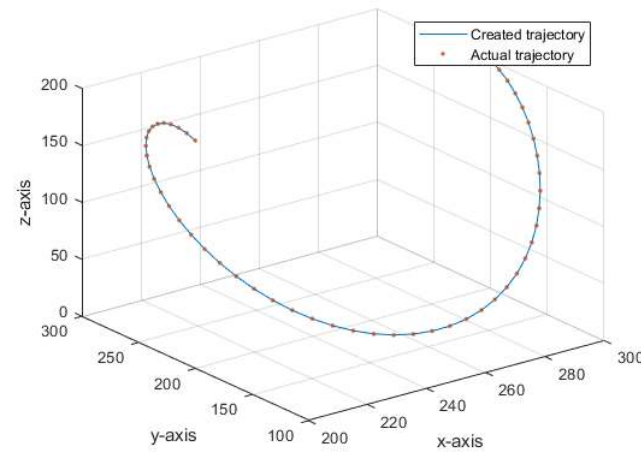
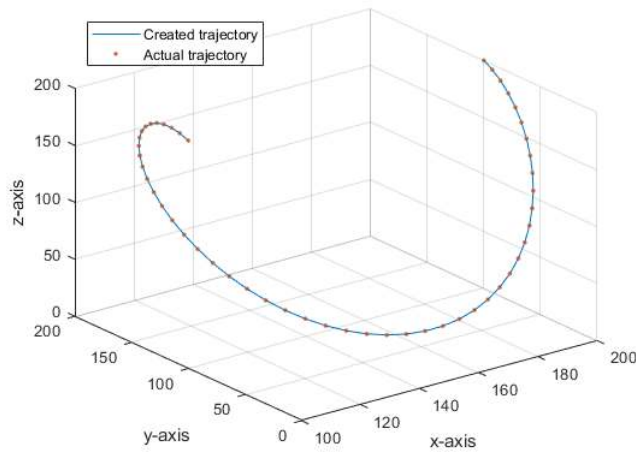
- ◆ Results for circle trajectory with 45 degrees diagonally downward



Experiments and Results

➤ Trajectories

- ◆ Results for sine trajectory with gripper downward



Experiments and Results

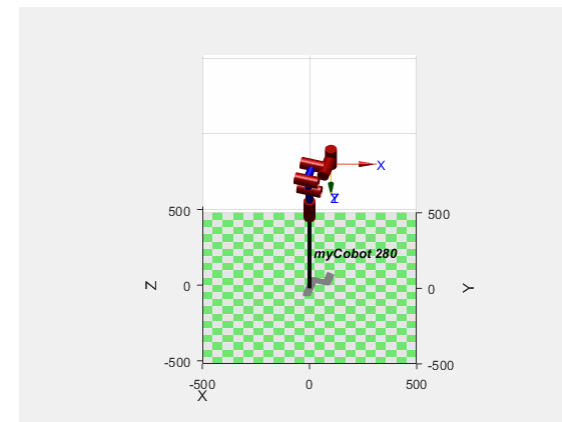
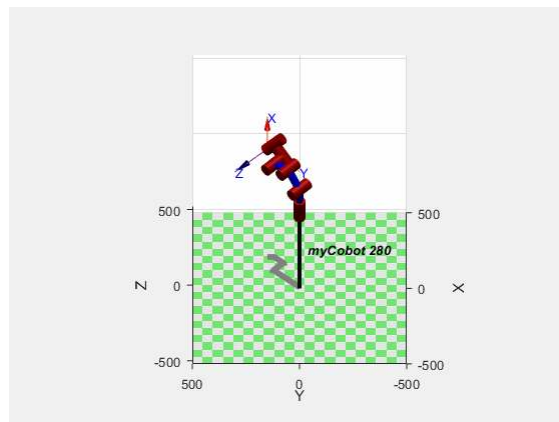
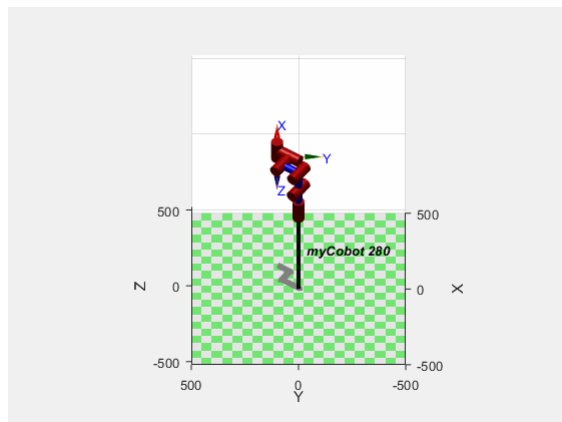
➤ Simulation

◆ Trajectories Circle

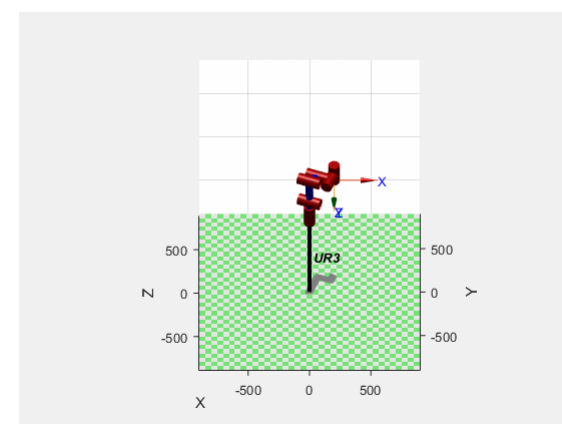
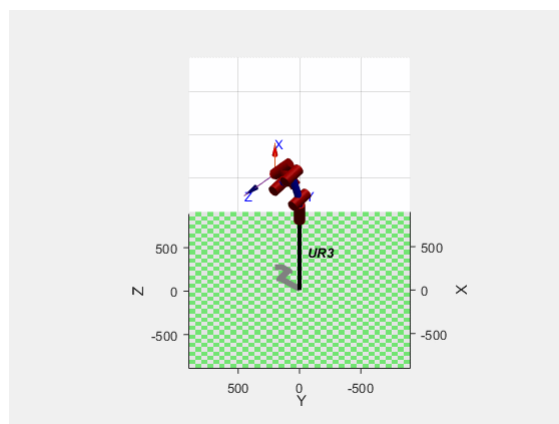
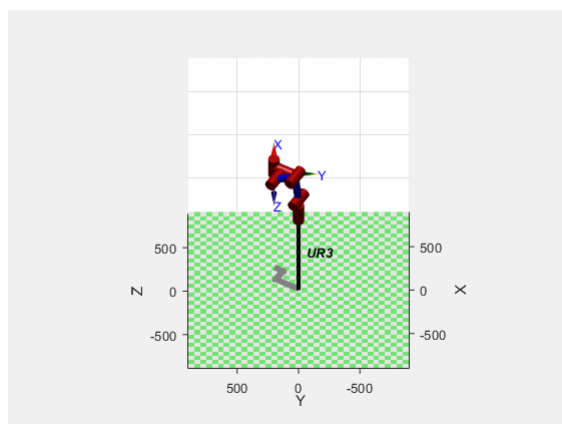
Circle/

Sine

myCobot



UR3



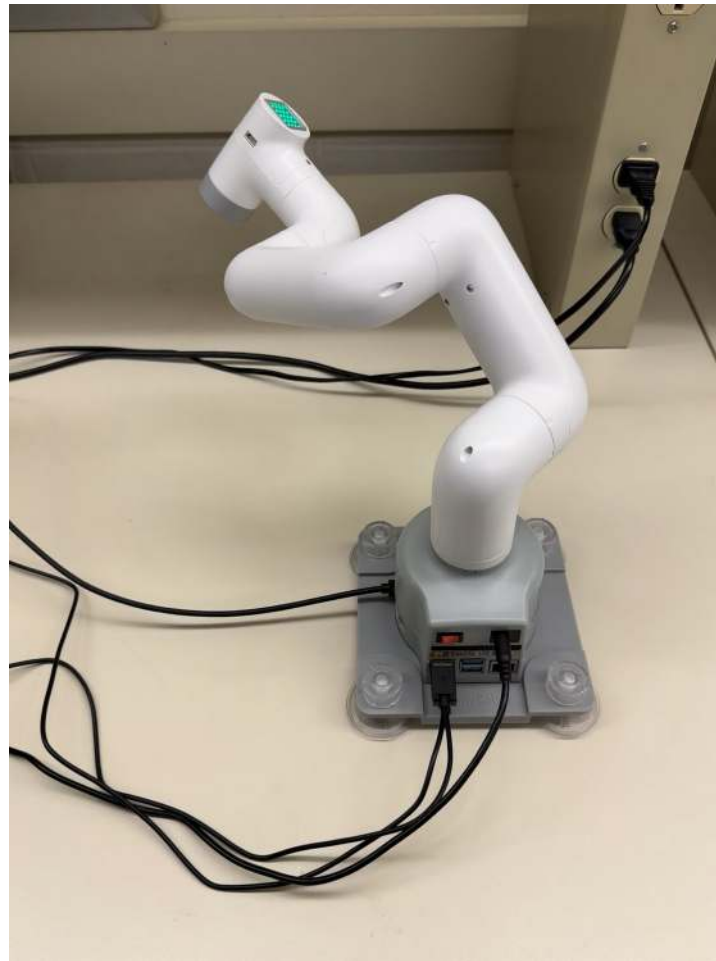
Experiments and Results

➤ Real-world implementation

◆ Trajectories Circle



Circle/



Sine



Conclusion and Future Work

CONCLUSION

- Addressed inverse kinematics problem with joint configuration of robotic arm using multi-objective optimization algorithm.
- While meeting the position and orientation error requirements, joint rotation is minimized.

FUTURE WORK

- Improve the third objective function to consider the dynamics in robotics
- Extend the study to include various and flexible robotic arm structures, such as higher DOF redundant robotic arms, and heterogeneous robots.
- Enhance motor control of the robotic arm using PID control or the fractional-order PID (FOPID) control with reinforcement learning for smoother robot control.

END

Thank you!

Questions & Comments